

BERKAY KULLUKÇU

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EDUCATION

Doctor of Engineering (Dr.-Ing.) in Electrical Engineering Technische Universität Dresden, Chair of Acoustics and Haptics, Dresden, Germany.	2023 - Present
Master of Science (MSc) in Mechanical Engineering Koç University, College of Engineering, Istanbul, Turkey.	2020 - 2022
Bachelor of Science (BSc) in Mechanical Engineering Koç University, College of Engineering, Istanbul, Turkey.	2015 - 2020

SKILLS AND INTERESTS

Acoustics & Psychoacoustics: Electric trucks, vehicle acoustics, AVAS, sound quality, semantic descriptor modeling, psychoacoustic metrics (roughness, tonality, sharpness), field studies, and project proposal writing (DFG).

Signal Processing & Simulation: SQL, MATLAB, Python, COMSOL FEM, Artemis Suite, data science, statistics, machine learning, and object-oriented programming.

Measurement & Experimentation: Free-field microphones, accelerometers, Squadriga II/III, binaural recording, listening tests, and perceptual modeling.

Languages: English (C1), German (C1).

RESEARCH EXPERIENCE

Department of Machine Dynamics and Acoustics (Scientific Employee) (TH Wildau) - Wildau, Germany	October 2025 - Present
Conducting research on acoustic- and vibration-based condition monitoring of rail-vehicle components, with a focus on detecting loose bolted joints and structural defects.	
Chair of Acoustics and Haptics (Dr.-Ing.) (AHA) - Dresden, Germany	April 2023 - Present
Research on AVAS sound design for electric trucks, including psychoacoustic testing, semantic evaluation, and binaural sound synthesis. Experiments address user comfort and the perceptual urgency of vehicle warning sounds.	
Bio-Integrated Microdevices Laboratory (Master's) (BMDL) - Istanbul, Turkey	June 2020 - September 2022
Research on flexible polymer energy harvesters, PMUTs, nano/microfabrication, and acoustic equivalent-circuit models.	
Bio-Integrated Microdevices Laboratory (Undergraduate) (BMDL) - Istanbul, Turkey	January 2020 - June 2020
Research on wearable sensor-system design for venous dislodgement detection in hospital patients.	

TEACHING EXPERIENCE

TH Wildau - Wildau, Germany	Oct. 2025 - Jan. 2026
Teaching-assistant duties in the senior engineering class IDM (Interdisciplinary Module).	
TU Dresden - Dresden, Germany	June 2024 - Jan. 2025
Supervised a master-level student during a six-month vehicle-acoustics research internship (final grade: 1.0).	
TU Dresden - Dresden, Germany	September 2023 - Present
- Sound Design lecture: supervision of senior engineering students in a vehicle sound-design project. - Acoustic Hands-on Experimenting laboratory: supervision of household-appliance sound-power measurement projects.	
Koç University - Istanbul, Turkey	September 2020 - Jan. 2021
MECH401 Mechanical Engineering Laboratory: tutoring for senior engineering students.	
The Educational Volunteers Foundation of Turkey (TEGV) - Istanbul, Turkey	Sep. 2018 - Jan. 2019
Programming tutor for middle-school students.	
Private Mathematics Teacher - Istanbul, Turkey	2017 - 2020
Private mathematics tutoring for primary- and middle-school students.	

INDUSTRIAL EXPERIENCE

GENERAL ELECTRIC (GE) AEROSPACE - Kocaeli, Turkey

Nov. 2022 - March 2023

Design Engineer

- Design work on CT7/T700 turboshaft and turboprop engines.

TEDAVEM Information Systems (LLC) - Istanbul, Turkey

June 2021 - Jan. 2023

Co-Founder

- Co-founded a hospital-digitalization start-up developing IoT systems for monitoring ventilation and infusion machines in patient rooms; designed integrated circuits for device communication.

KORDSA - Istanbul, Turkey

March 2022 - September 2022

Flexible Electronics Researcher / R&D

- Worked in the Thin Films & Flexible Electronic Sensors Department (0.35% applicant acceptance rate).
- Developed LabVIEW software for testing.

TURKISH AEROSPACE (TAI) - Ankara, Turkey

November 2019 - May 2020

Mechanical Engineering Intern

- Worked in the Safety Management Systems Department.

GROUPE RENAULT - Bursa, Turkey

June 2019 - August 2019

Automotive Engineering Intern / R&D

- Developed software exceeding 20,000 lines of code to help R&D engineers manipulate CATIA 3D parts.

TURKISH AEROSPACE (TAI) - Istanbul, Turkey

November 2019 - January 2020

Mechanical Engineering Intern / R&D

- Performed Patran FEM and 3D CAD work for aerospace component design.

PUBLICATIONS

1. B. Kullukçu, J. Krautwurm, S. Merchel, R. Rosenkranz, and M. E. Altinsoy, "Investigating perceptual discrimination thresholds for attributes of whole-body vibration," *Scientific Reports*, 2026. - [Link](#)
2. B. Kullukçu, R. Rosenkranz, and M. E. Altinsoy, "Electric Truck Sound Design: A User-Centered Psychoacoustic Approach to AVAS," *Journal of the Audio Engineering Society*, 2025. - [Link](#)
3. B. Kullukçu et al., "Piezoelectric PVDF-TrFE/PET Energy Harvesters for Structural Health Monitoring (SHM) Applications," *Integrated Ferroelectrics*, 2023. - [Link](#)
4. B. Kullukçu and L. Beker, "Design and Modeling of a PVDF-TrFE Flexible Wind Energy Harvester," *Turkish Journal of Electrical Engineering and Computer Sciences*, 2023. - [Link](#)
5. Piezoelectric PVDF-TrFE/PET Energy Harvesters for Structural Health Monitoring (SHM) Applications in Wind Turbines - Master's Thesis, 2023. - [Link](#)

CONFERENCES

1. B. Kullukçu, S. Atamer, J. Krautwurm, and M. E. Altinsoy, "Perceived Annoyance in Multi-Source Electric Vehicle AVAS Environments," *ICSV32*, Istanbul, Turkey, 2026. - [Proceedings](#)
2. B. Kullukçu, R. Pianowski, and D. Hannebauer, "Frequency-Modulated and Single-Tone Excitation to Reveal Vibro-Acoustic Nonlinearities in Loosened Bolted Joints," *Forum Acusticum*, Graz, Austria, 2026. - [Link](#)
3. B. Kullukçu, R. Kamenzky, and D. Hannebauer, "SIAMIS: Experimental Study on the Acoustic Behavior of Bolted Joints Under Varying Tightening Torques," *Proc. DAGA*, Dresden, Germany, 2026. - [Link](#)
4. B. Kullukçu, M. Engel, S. Atamer, and M. E. Altinsoy, "Psychoacoustic Modelling of AVAS Sounds: Consumer-Centric Semantic Attribute Development for Electric Vehicles," *Proc. Forum Acusticum*, Málaga, Spain, 2025. - [Link](#)
5. B. Kullukçu, O. Zeidler, S. Atamer, and M. E. Altinsoy, "Experimental Determination of the Masking Threshold in Electric Trucks," *Proc. DAGA*, Copenhagen, Denmark, 2025. - [Link](#)
6. B. Kullukçu, J. W. Lee, S. Atamer, and M. E. Altinsoy, "Designing Effective Warning Sounds: Psychoacoustical Considerations," *Proc. INTER-NOISE*, Nantes, France, 2024. - [Link](#)
7. B. Kullukçu, R. Rosenkranz, and M. E. Altinsoy, "Acoustic Vehicle Alerting Systems (AVAS) for Electric Trucks: Initial Thoughts for the Requirements," *Proc. DAGA*, Hannover, Germany, 2024. - [Link](#)
8. P. T. Ziarati, B. Kullukçu, and L. Beker, "Modeling miniaturized piezoelectric ultrasound transducers: comparison of lumped and finite element models," *2020 IEEE International Ultrasonics Symposium*, Las Vegas, NV, USA, 2020. - [Link](#)

PATENTS

1. **B. Kullukçu, G. Umut, and B. Ataseven, “Momentum Nasogastric Tube System,” TR2022001302A1, Oct. 21, 2024. - [Link](#)**
2. **B. Kullukçu, G. Umut, and B. Ataseven, “An Invasive Device,” TR2021015887A1, May 21, 2024. - [Link](#)**
3. **B. Kullukçu, G. Umut, and B. Ataseven, “Detecting Device and System for Infusion Pumps,” Utility Model TR2022007637Y, Apr. 22, 2024. - [Link](#)**

SCHOLARSHIPS & AWARDS

1. Travel Grant for ICSV32 (Förderprogramm Kongressreisen 2026) - DAAD
2. EAA Travel Grant for Forum Acusticum 2025 - European Acoustics Association
3. V. Patent Day (2019-2021) Patent Holder Award - Koç University
4. BİDEB 2210-C National Scholarship Program for MSc Students (Priority Research Areas) - TÜBİTAK
5. TEYDEB 1512 - USD 25,000 Techno-Startup Funding - TÜBİTAK
6. Design Innovation Challenge, 1st Prize - Undergraduate & Graduate Scholarship - Garip & Zeycan Foundation
7. Graduate Academy Conference Travel Grant for DAGA 2025 Copenhagen - TU Dresden
8. VI. Patent Day Patent Holder Award - Koç University

SESSION CHAIR DUTIES

1. “Körperschall” (Structure-Borne Sound) Session - DAGA 2026, Dresden, Germany - [Link](#)
2. T14 SS03: Sound Quality and Product Design - 32nd International Conference on Sound and Vibration (ICSV32) 2026, Istanbul, Turkey - [Link](#)